CS6741 — Statistical Foundations of Data Science + related stats/optimisation

Tier	C
Phase (order)	7
Do this first	CS6015, CS4011
Estimated effort	12 weeks
Unlocks	CS6464, CS6730, CS7370

Roadmap

1. Estimation, hypothesis testing, regression theory; then PGMs and optimisation.

Video courses (all links checked on YouTube on 2026-10-22)

🎓 Main course: follow it start to end

- Statistics 110: Probability (35 videos). Harvard *Stat 110: Probability* (Joe Blitzstein, official Harvard channel): the best probability course online. Free book at probabilitybook.net.

🎥 Other full lecture series (pick one if the main one doesn't click, or for topics it skips)

- NOC Jan 2019 : Statistical Inference (65 videos). NPTEL *Statistical Inference* (IIT Kharagpur): estimation, MLE, hypothesis tests.
- Statistical Learning with Python (100+ videos). Stanford *Statistical Learning with Python* (Hastie & Tibshirani, official): the ISLP book as lectures.
- Stanford EE364A Convex Optimization I Stephen Boyd I 2023 (18 videos). Stanford EE364A *Convex Optimization* (Stephen Boyd, 2023, official): for the optimisation half.

Study plan (week by week)

Lecture numbers are positions in the main playlist above. Click a number to open that video.

Week	Lectures	What to do
1	1–6	Counting, axioms, conditional probability, Bayes, Monty Hall. Do Stat 110 strategic practice 1-2 (projects.iq.harvard.edu/stat110).
2	7–11	Random variables, Bernoulli/binomial/hypergeometric, expectation, indicators, Poisson.
3	12–14, 16	Continuous distributions: uniform, normal, exponential; LOTUS. Simulate each in NumPy and compare histograms with the PDFs.

Week	Lectures	What to do
4	17–20	Moment generating functions, joint/marginal/conditional distributions, multinomial.
5	21–24	Covariance, correlation, transformations, convolutions, Beta and Gamma, Poisson process.
6	25–27	Order statistics, conditional expectation (Adam's and Eve's laws).
7	28–30	Inequalities, law of large numbers, central limit theorem, chi-square/t, multivariate normal. Demonstrate the CLT by simulation.
8	31–34	Markov chains, stationary distributions. Simulate a random walk on a graph and compare with the stationary distribution.

▼ **All lectures in *Statistics 110: Probability* (35 videos)**

1. Lecture 1: Probability and Counting | Statistics 110
2. Lecture 2: Story Proofs, Axioms of Probability | Statistics 110
3. Lecture 3: Birthday Problem, Properties of Probability | Statistics 110
4. Lecture 4: Conditional Probability | Statistics 110
5. Lecture 5: Conditioning Continued, Law of Total Probability | Statistics 110
6. Lecture 6: Monty Hall, Simpson's Paradox | Statistics 110
7. Lecture 7: Gambler's Ruin and Random Variables | Statistics 110
8. Lecture 8: Random Variables and Their Distributions | Statistics 110
9. Lecture 9: Expectation, Indicator Random Variables, Linearity | Statistics 110
10. Lecture 10: Expectation Continued | Statistics 110
11. Lecture 11: The Poisson distribution | Statistics 110
12. Lecture 12: Discrete vs. Continuous, the Uniform | Statistics 110
13. Lecture 13: Normal distribution | Statistics 110
14. Lecture 14: Location, Scale, and LOTUS | Statistics 110
15. Lecture 15: Midterm Review | Statistics 110
16. Lecture 16: Exponential Distribution | Statistics 110
17. Lecture 17: Moment Generating Functions | Statistics 110
18. Lecture 18: MGFs Continued | Statistics 110
19. Lecture 19: Joint, Conditional, and Marginal Distributions | Statistics 110
20. Lecture 20: Multinomial and Cauchy | Statistics 110
21. Lecture 21: Covariance and Correlation | Statistics 110
22. Lecture 22: Transformations and Convolutions | Statistics 110
23. Lecture 23: Beta distribution | Statistics 110
24. Lecture 24: Gamma distribution and Poisson process | Statistics 110
25. Lecture 25: Order Statistics and Conditional Expectation | Statistics 110
26. Lecture 26: Conditional Expectation Continued | Statistics 110
27. Lecture 27: Conditional Expectation given an R.V. | Statistics 110
28. Lecture 28: Inequalities | Statistics 110

29. Lecture 29: Law of Large Numbers and Central Limit Theorem | Statistics 110
30. Lecture 30: Chi-Square, Student-t, Multivariate Normal | Statistics 110
31. Lecture 31: Markov Chains | Statistics 110
32. Lecture 32: Markov Chains Continued | Statistics 110
33. Lecture 33: Markov Chains Continued Further | Statistics 110
34. Lecture 34: A Look Ahead | Statistics 110
35. Joseph Blitzstein: "The Soul of Statistics" | Harvard Thinks Big 4

Books, notes and websites

- Book: Casella & Berger, *Statistical Inference*
- Book: Boyd & Vandenberghe, *Convex Optimization* (free) + Stanford EE364a
- Book: Koller & Friedman, *Probabilistic Graphical Models*

Practice / proof of understanding

Fit a Bayesian network and a logistic model, test both, and report confidence intervals.

What your IITM course sheet says

From the PDF: CS6741

Official description: This course provides an introduction to statistics as applied to data science. It is intended as a first course for students wanting to specialise in the fields of Data Science and Artificial Intelligence. Upon completion of the program, students should: ? Understand the role of statistics in data science, and contextualise it in the broader domain of machine learning. ? Interpret different data visualisation methods and choose the right visualisation for a specific data representation task. ? Know how to compute statistical measures of centrality and dispersion. ? Demonstrate knowledge of point and interval estimators on fixed-sample and large-sample data. ? Distinguish between Type I and Type II errors ? Understand the central role of hypothesis testing and perform different tests on mean, variance, and proportions under different conditions ? Know how to carefully design e

Official syllabus (trimmed):

> Descriptive statistics: [2 weeks] Types of data such as qualitative[nominal, ordinal], quantitative[discrete, continuous]. Data visualisation tools such as relative, cumulative, absolute frequency charts and tables, relative, cumulative absolute histograms and frequency tables, scatter plots. Summary statistics such as mean, median, mode, percentiles, range, inter quartile range, variance, standard deviation, covariance, correlation (Pearson, Spearman rank correlation. Point Biserial correlation) Probability Theory (Recap): [2 weeks] Sample space, events, axioms of probability, random variables, expectation, joint distributions, condition distributions. Discrete probability distributions such as Bernoulli, Binomial, Uniform and Poisson. Continuous Distributions such as uniform distribution, normal distribution, exponential family. Analysis of multivariate Gaussian distribution. Randomised control experiments: [0.5 week] Sampling strategies such as simple random sampling, convenience sampling, stratified sampling and cluster sampling. Designing experiments with an understanding of explanatory variables, response variables, lurking variables, treatments, control groups, experimental units, placebo, single blind and double blind experiments. Distributions of sampling statistics: [1 week] Central limit theorem, distribution of mean, variance and proportions Estimation: [2.5 weeks]

Official textbooks: 1) Ross, Sheldon M. Introductory statistics. Academic Press, 2017 2) Computer Age Statistical Inference: Algorithms, Evidence and Data Science, Bradley Efron and Trevor Hastie. 2016 (Cambridge University Press).

Official references: 1. The Elements of Statistical Learning: Data Mining, Inference, and Prediction. 2nd Edition, Trevor Hastie, Robert Tibshirani, and Jerome Friedman, 2009 (Springer) [Available online: https://web.stanford.edu/~hastie/ElemStatLearn/] 2. Casella, George and Berger Roger L. Statistical Inference. Cengage Learning, 2001

From the PDF: CS6730

Official description: The course provides comprehensive introduction to probabilistic graphical models. At the end of the course the student should be able to model problems using graphical models; design inference algorithms; and learn the structure of the graphical model from data.

Official syllabus (trimmed):

> Review: Fundamentals of Probability Theory - Views of Probability, Random Variables and Joint Distributions, Conditional Probability, Conditional Independence, Expectation and Variance, Probability Distributions - Conjugate Priors, Introduction to Exponential Family; Fundamentals of Graph Theory - Paths, Cliques, Subgraphs, Cycles and Loops. Graphical Models: Introduction - Directed Models (Bayesian Network), Undirected Models (Markov Random Fields), Dynamic Models (Hidden Markov Model & Kalman Filters) and Factor Graph; Conditional Independence (Bayes Ball Theorem and D-separation), Markov Blanket, Factorization (Hammersley-Clifford Theorem), Equivalence (I-Maps & Perfect Maps); Factor Graphs - Representation, Relation to Bayesian Network and Markov Random Field. Inference in graphical models: Exact Inference - Variable Elimination, Elimination Orderings, Relation to Dynamic Programming, Dealing with Evidence, Forward-Backward Algorithm, Viterbi Algorithm; Junction Tree Algorithm; Belief Propagation (Sum Product); Approximate Inference - Variational Methods (Mean Field, Kikuchi & Bethe Approximation), Expectation Propagation, Gaussian Belief Propagation; MAP Inference - Max-Product, Graph Cuts, Linear Programming Relaxations to MAP (Tree-Reweighted Belief Propagation, MPLP); Sampling - Markov Chain Monte Carlo, Metropolis Hastings, Gibbs (Collapsing & Blocking),

Official textbooks: Koller, D. and Friedman, N. (2009). Probabilistic Graphical Models: Principles and Techniques. MIT Press.

Official references: 1. Jensen, F. V. and Nielsen, T. D. (2002). Bayesian Networks and Decision Graphs. Information Science and Statistics. Springer, 2nd edition. 2. Kevin P. Murphy (2013) Machine Learning: A Probabilistic Perspective. 4th Printing. MIT Press. 3. Barber, D. (2011). Bayesian Reasoning and Machine Learning. Cambridge University Press, 1st edition. 4. Bishop, C. M. (2011). Pattern Recognition and Mac

From the PDF: CS5020

Official description: Optimisation problems arise in a variety of disciplines, for e.g., machine learning, reinforcement learning, signal processing, networks. This course covers the core concepts of continuous optimisation and includes a thorough introduction to unconstrained and constrained optimisation problems. Popular algorithms such as gradient and Newton methods will be introduced and analysed for both convex and non-convex optimisation settings. The evaluation will be based on the exams and home-works(which will involve both theoretical and programming components).

Official syllabus (trimmed):

> Part 0: Recap of real analysis and linear algebra (Lectures 1-3) 1. Sets, functions, sequences, continuity, differentiability, gradients, Taylor series expansion. 2. Vectors, matrices, norms, symmetric matrices, eigenvalue decomposition, positive semidefinite and positive definite matrices. Part I: Convex sets and functions (Lectures 4 – 11) 1. Convex sets, examples and properties. 2. Convex functions, strict and strong convexity, examples, and convexity preserving operations. 3. Equivalent definitions of convexity under differentiability assumptions. Part II: Unconstrained optimisation (Lectures 12-20) 1. Maxima, minima, stationary point, saddle point, local and global maximum/minimum. 2. First order and second order conditions for optimality. 3. Linear, quadratic and convex optimisation problems, examples. 4. Benefits of convexity. Part III: Constrained optimisation (Lectures 21-29) 1. Constrained optimisation problem, feasible set. 2. Lagrangian, KKT conditions 3. Linear and quadratic optimisation 4. Duality for convex optimisation — theorem of alternatives, Farka's lemma. Part IV: Algorithms for optimisation (Lectures 30-40) 1. Gradient descent with fixed step size, line search and Armijo-Goldstein rule 2. Newton method and variations 3. Conjugate gradient and Quasi-newton methods. 4. Algorithms for constrained optimisation: Projected gradient descent,

Official textbooks: 1. Stephen Boyd, and Lieven Vandenberghe. Convex optimization. Cambridge university press, 2004. 2. David G. Luenberger and Yinyu Ye. Linear and nonlinear programming. 4th edition. Springer, 2015.

Official references: 1. Dimitri P. Bertsekas. Nonlinear programming. Belmont: Athena scientific, 1999. 2. Jorge Nocedal and Stephen Wright. Numerical optimization. Springer, 1999